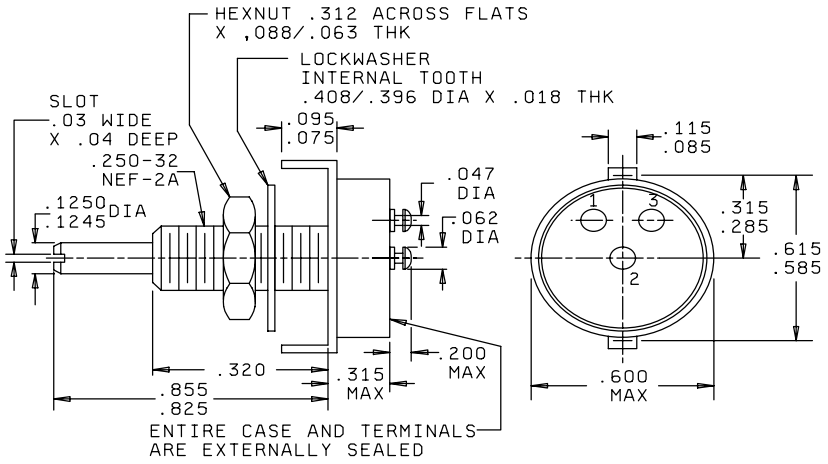
INCH-POUND
MIL-PRF-39002/3A
1 February 1999
SUPERSEDING
MIL-R-39002/3
31 March 1980

PERFORMANCE SPECIFICATION SHEET

RESISTOR, VARIABLE, WIRE-WOUND, SEMI-PRECISION, STYLE RK11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-39002.



Inches	mm	Inches	mm	Inches	mm
.018	.46	.095	2.41	.396	10.06
.03	.8	.115	2.92	.408	10.36
.04	1.0	.1245	3.162	.585	14.86
.047	1.19	.1250	3.175	.600	15.24
.062	1.57	.250	6.35	.615	15.62
.063	1.60	.285	7.24	.825	20.96
.075	1.90	.312	7.92	.855	21.72
.085	2.16	.315	8.00	4.000	101.60
.088	2.24	.320	8.13		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style RK11 resistor.

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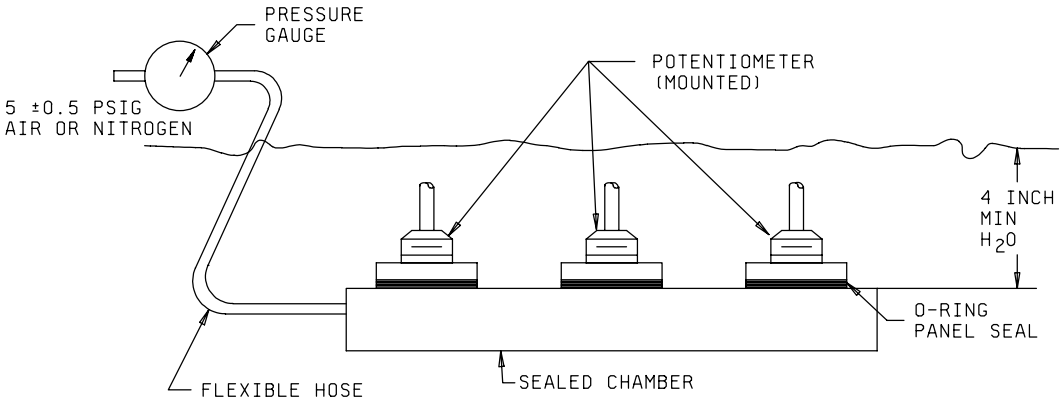

FIGURE 2. Shaft to bushing seal test.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table 1.

Resistance value: 5,000 ohms ±5 percent.

Shaft length: See table I.

Shaft style: See table I.

Bushing: See table I.

Power rating: 1.5 watts at full load operation at an ambient temperature of 85°C.

Nominal resolution: See table I.

Rated working voltage: See table I.

Hardware: Hardware shall be as specified in figure 1 and shall be corrosion resistant. (Hardware optional between supplier and prime contractor.)

Mechanical travel: 360 degrees.

Actual effective-electrical travel: 300 degrees ±3 degrees.

Linearity: Independent linearity ±1 percent.

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TABLE I. Style RK09 resistor.

Type designation 1/	Shaft length inches ±.015 shaft style	Nominal total resistor value (ohms)	Nominal resolution (percent) 2/	Rated ac and dc working voltage (volts)	Terminal	Dash number
RK11SLCS502	.840 inch slotted	5,000	0.17	122	S	0001

1/ For reference purposes only.

2/ Percent of resolution = $\frac{1}{N}$ x 100. N = number of turns.

Dielectric withstanding voltage:

At atmospheric pressure: Method 301 of MIL-STD-202, test voltage 900 volts rms.

At reduced barometric pressure: Method 105 of MIL-STD-202, test condition D.

Electrical output: The electrical output (in terms of percent of applied voltage) is linear with respect to the angular position of the contact arm.

Insulation resistance: Method 302 of MIL-STD-202, test condition A.

Torque: The torque required to effect rotation shall be 0.5 ounce-inches to 6 ounce-inches.

Thermal shock: Method 107 of MIL-STD-202, test condition B.

Resistance temperature characteristic: Method 304 of MIL-STD-202.

Moisture resistance: Method 106 of MIL-STD-202.

Shock, specified pulse: Method 205 of MIL-STD-202, test condition G.

Vibration high frequency: Method 204 of MIL-STD-202, test condition D.

Salt spray: Method 101 of MIL-STD-202, test condition A.

Resistance to soldering heat: Method 210 of MIL-STD-202, test condition A.

Life: Method 108 of MIL-STD-202, test condition D.

Rotational life: In accordance with MIL-R-39002 except the cycle angle shall be not less than 98 percent of the effective electrical travel angle. Electrical continuity shall be maintained throughout the cycle.

Terminal strength: Method 211 of MIL-STD-202, test condition A.

Strength of mounting bushing: The potentiometer shall withstand, without damage, a torque of from 15.5 inch-pounds to 16.5 inch-pounds applied to tighten the mounting nut for five seconds.

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Shaft-to-bushing seal (O-ring shaft seal):

Condition the sample at -62°C and +71°C for 15 minutes at each temperature extreme with five minutes maximum between temperature extremes and return to room temperature.

Mount the test units as shown in figure 2. The mounting nut shall be torqued to 15.5 inch pounds to 16.5 inch pounds.

Immerse the test chamber in water as shown in figure 2. Pressurize the sealed test chamber with dry air or nitrogen to 5 psig ±0.5 psig for 60 seconds ±10 seconds.

There shall be no evidence of a continuous stream of bubbles emanating from the shaft to bushing seal.

No failures are allowed.

Qualification: Those manufacturers qualified to style RK09 can extend qualifications to style RK11 by submitting and passing the following additional test: 13 units tested to the strength of mounting busing test, and 13 units to the shaft-to-bushing seal test. No failures are allowed.

Conformance inspection: The following test shall be included for group A inspection:

Shaft-to-bushing seal

Part or Identifying Number (PIN): Consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

M39002/03-0001

Basic document Number	Dash number from table I

Custodian:
Army - CR

Review activity:
Army - AR

Preparing activity:
DLA - CC

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